

Provider Document db Process

- [Introduction](#)
 - [Decision Log](#)
- [Provider Inbound High Level Flow Diagram](#)
 - [Lambda Flow diagram](#)
 - [lambda processing](#)
 - [Glue Flow diagram](#)
 - [Initial Processing](#)
 - [Mapping to scdhhs canonical model](#)
 - [Document DB Insertion](#)
 - [Final Process](#)
 - [Glue Configuration](#)
 - [Spark SQL](#)
 - [CLIA spark SQL](#)
 - [EFT SQL](#)
 - [MMIS Provider SQL](#)
 - [NPI Crosswalk SQL](#)
 - [NPI Junction SQL](#)
 - [Auxiliary SQL](#)
 - [Affiliation SQL](#)
 - [Primary Tin SQL](#)
 - [Scies Vendor SQL](#)
 - [Provider canonical structure](#)
 - [Collection names](#)
 - [Parameter store Details](#)
 - [Secret manager configuration](#)
 - [S3 Bucket configuration](#)
 - [Email Notifications](#)
 - [Lambda Notifications](#)
 - [Glue Job Notifications](#)
- [Notes](#)

Introduction

As part of provider refactoring process following are the changes

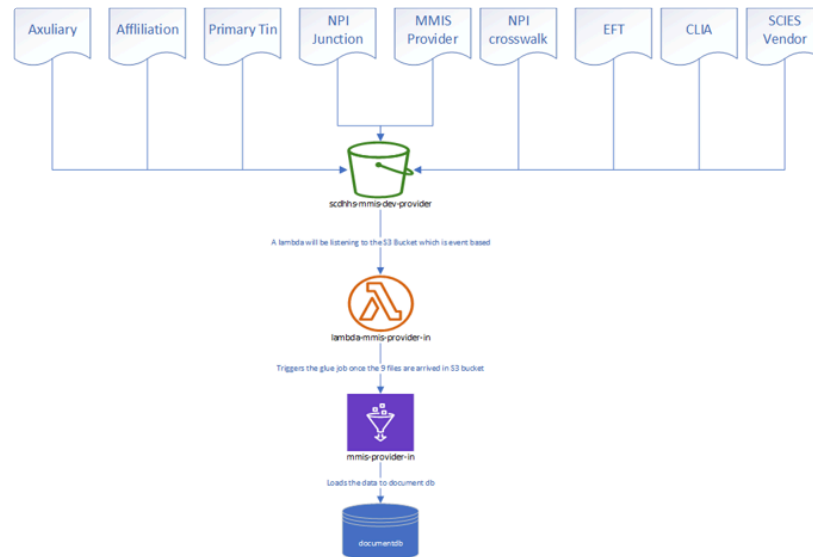
- All the harmonization logic and mappings will be done in Glue
- The scdhhs canonical model will be loaded to Document DB
- There will be some changes to the process and attributes as per the standards

Please refer to architecture diagram

Decision Log

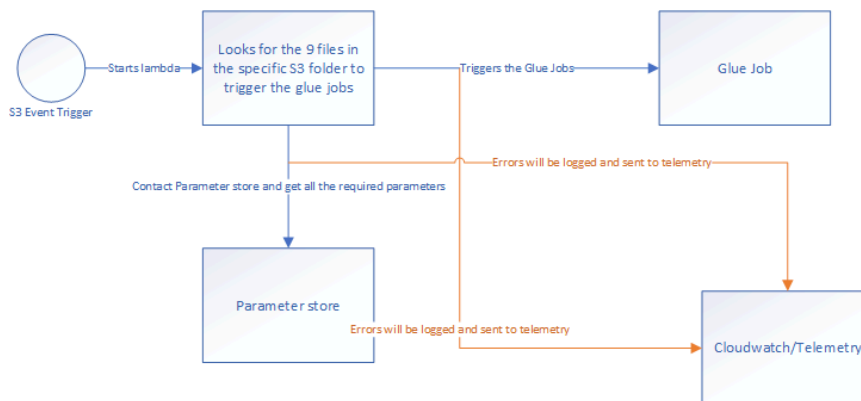
- As per Architecture decision there no concept of RDL
- As per Architecture decision we need to overwrite the documents in ODS.

Provider Inbound High Level Flow Diagram



Lambda Flow diagram

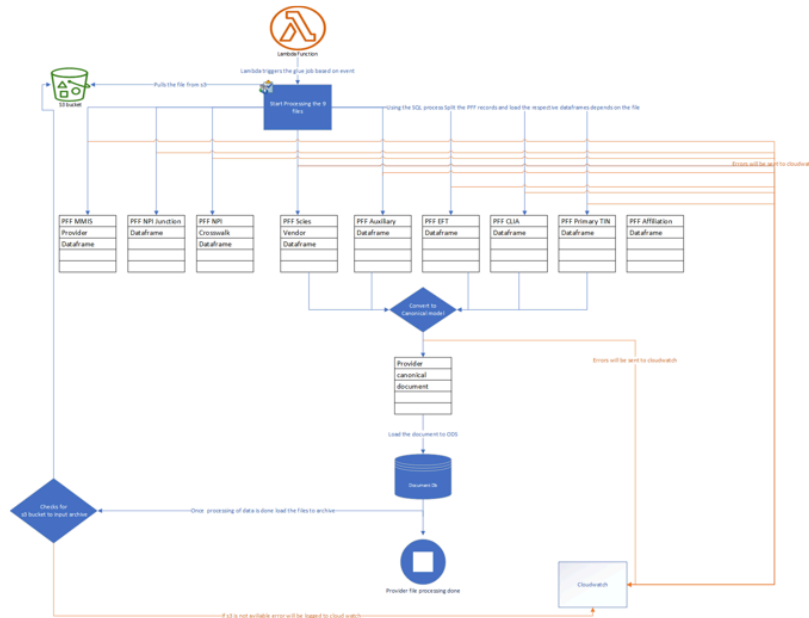
Lambda will trigger Glue job once all the four files arrived in the S3 Bucket



lambda processing

- Lambda will look for the specific folder in the s3 bucket to trigger the glue job by pulling the parameters for the parameter store.
- Once the 9 files (MMIS provider, NPIJunction, ProviderAuxiliary, ProviderAffiliation, ProviderEFT, ProviderCLIA, Scies Vendor, NPI Crosswalk, Primary Tin) are dropped to the respective folders in S3 bucket the glue job will be triggered for further processing of provider records.
- Any errors in the process will be logged to cloud watch.

Glue Flow diagram



Initial Processing

- Lambda will trigger the glue job.
- Once the glue job is triggered the files (MMIS provider, NPIJunction, ProviderAuxiliary, ProviderAffiliation, ProviderEFT, ProviderCLIA, Scies Vendor, NPI Crosswalk, Primary Tin) will be pulled from the S3 for further processing.
- The MMIS provider, NPIJunction, ProviderAuxiliary, ProviderAffiliation, ProviderEFT, ProviderCLIA, Scies Vendor, NPI Crosswalk, Primary Tin data PFF records will be loaded to the respective data frames by using the SQL configuration.

Mapping to scdhhs canonical model

- The respective theMMIS provider, NPIJunction, ProviderAuxiliary, ProviderAffiliation, ProviderEFT, ProviderCLIA, Scies Vendor, NPI Crosswalk, Primary Tin data PFF records data will be mapped to scdhhs canonical model.
- Metadata will be added to the mapped scdhhs canonical provider Json and will be further processed.
- Any Processing errors in the mapping will be logged to cloud watch and the raw record will be loaded to provider error collections.

Document DB Insertion

- The mapped final Json will be loaded to ODS to the respective final collection if the connection to document db is success.
- Any connection errors to document db will be logged to cloud watch.

Final Process

- Once the process is successful the file will be archived to the archive folder in S3

Glue Configuration

Glue Job name in lower Environment: mmis-provider-in

Under Basic properties

Please provide the following properties

Name: Provide the name of the glue (in lower mmis-provider-in)

IAM Role: Provide the Iam role created for glue(in lower scdhhs-mes-glue-service-role-lower)

Type: Spark

Glue version: Spark2.0 -Supports spark 2.4, Scala 2, Python 3

Language: Python 3

Worker Type: G1X

Requested number of workers: 10

Select Generate Job insights.

Number of retries:0

Job timeout:2880

Under Advanced properties

Please provide the following properties

Script File name: mmis-babynet-in.py

Script path: <s3://scdhhs-mmis-dev-lower/dev/code/src/>

Max Concurrency: 2

Temporary Path: <s3://scdhhs-mmis-dev-lower/dev/code/temp>

Under connections

Please provide the following properties

Chose the connection depending on the environment: scdhhs-mes-blue

Under Libraries

Please provide the following properties

Dependent Jar paths : <s3://scdhhs-mmis-member-lower/dev/code/lib/commonPythonlib.zip>,<s3://scdhhs-mmis-member-lower/dev/code/lib/babynetLibrary.zip>

Job parameters

Please provide the following properties

key:**parameterName** <value:/mmis/lower/provider/in>

key:**secretKeyName** value:scdhhs-mes-blue/docdb

Tags:

key:env value:lower

Spark SQL

CLIA spark SQL

```
1 get_clia_provider_sql_query(): sqlcmd = u''' select named_struct( "CLIA-PROV-NUMBER", trim(substring(value, 1, 6)), "CLIA-ID-OF-LAST-UPDT", trim(substring(value, 7, 8)), "CLIA-DT-LAST-UPDATE", named_struct( "CLIA-DLU-CN", trim(substring(value, 15, 2)), "CLIA-DLU-YMD", named_struct( "CLIA-DLU-YR", trim(substring(value, 17, 2)), "CLIA-DLU-MO", trim(substring(value, 19, 2)), "CLIA-DLU-DA", trim(substring(value, 21, 2)) ) ), "CLIA-ID-NUMBER", trim(substring(value, 23, 10)), "CLIA-CERT-INFO", transform(split(trim(TRAILING "~" FROM regexp_replace(trim(substring(value, 33, (17 * 2))), "{.17}","$1~")), '~'), value -> named_struct( "CLIA-CERTIFICATION-TYPE", trim(substring(value, 1, 1)), "CLIA-EFF-DATE", named_struct( "CLIA-EFF-CN", trim(substring(value, 2, 2)), "CLIA-EFF-YMD", named_struct( "CLIA-EFF-YR", trim(substring(value, 4, 2)), "CLIA-EFF-MO", trim(substring(value, 6, 2)), "CLIA-EFF-DA", trim(substring(value, 8, 2)) ) ), "CLIA-EXP-DATE", named_struct( "CLIA-EXP-CN", trim(substring(value, 10, 2)), "CLIA-EXP-YMD", named_struct( "CLIA-EXP-YR", trim(substring(value, 12, 2)), "CLIA-EXP-MO", trim(substring(value, 14, 2)), "CLIA-EXP-DA", trim(substring(value, 16, 2)) ) ) ), "CLIA-INDICATOR", trim(substring(value, 67, 1)) ), "info", named_struct( "title", "provider", "version", "0.0.1", "baseUri", "http://scdhhs.gov/provider/" ) ) as instance, concat("/kafka/",md5(value), "/", uuid(), ".json") as fullurl from ( select value , array_repeat('a', 0) as EMPTY_ARRAY from fixedLengthInput ) '''RETURN sqlcmd
```

EFT SQL

```
1 get_eft_prov_record_sql_query(): sqlcmd = u''' select named_struct( "PROV-NUMBER", trim(substring(value, 1, 10)), "EFT-MPI-KEY", named_struct( "EFT-NP-PRE", trim(substring(value, 1, 4)), "EFT-PROV-NUMBER", trim(substring(value, 5, 6)) ), "EFT-BANK-ID-NO", named_struct( "EFT-BANK-ROUTING-NO", int(translate(concat( if(instr('}jklmnopqr', substring(value, 11 + 8 - 1, 1)) > 0, '-', '' ), if(trim(LEADING '0' FROM substring(value, 11, if(-1 > -1, -1, 8))) = '', '0', '' ), trim(LEADING '0' FROM substring(value, 11, if(-1 > -1, -1, 8))) ), if(-1 > -1, '.', '' ), if(-1 > -1, substring(value, 11 + -1, 8 - -1), '' ) ), '{ABCDEFghi}jklmnopqr', '01234567890123456789')) ), "EFT-BANK-CHECK-DIGIT", int(translate(concat( if(instr('}jklmnopqr', substring(value, 19 + 1 - 1, 1)) > 0, '-', '' ), if(trim(LEADING '0' FROM substring(value, 19, if(-1 > -1, -1, 1))) = '', '0', '' ), trim(LEADING '0' FROM substring(value, 19, if(-1 > -1, -1, 1))) ), if(-1 > -1, '.', '' ), if(-1 > -1, substring(value, 19 + -1, 1 - -1), '' ) ), '{ABCDEFghi}jklmnopqr', '01234567890123456789')) ) ), "EFT-ACCOUNT-NUMBER", trim(substring(value, 20, 17)), "EFT-TYPE-OF-ACCOUNT", trim(substring(value, 37, 1)), "EFT-CONTACT", trim(substring(value, 38, 25)), "EFT-TELEPHONE", named_struct( "EFT-AREA-CODE", trim(substring(value, 63, 3)), "EFT-EXCHANGE", trim(substring(value, 66, 3)), "EFT-PHONE-NUMBER", trim(substring(value, 69, 4)) ), "EFT-STATUS-HISTORY-ARRAY", named_struct( "EFT-STATUS-HISTORY", transform(split(trim(TRAILING "~" FROM regexp_replace(trim(substring(value, 73, (12 * 6))), "{.12}","$1~")), '~'), value -> named_struct( "EFT-STATUS", trim(substring(value, 1, 1)), "EFT-ID-OF-LAST-UPDATE", trim(substring(value, 2, 5)), "EFT-DATE-OF-LAST-UPDATE", named_struct( "EFT-DLU-YR", trim(substring(value, 7, 2)), "EFT-DLU-MO", trim(substring(value, 9, 2)), "EFT-DLU-DA", trim(substring(value, 11, 2)) ) ) ), "EFT-REC-ID-LAST-UPDT", trim(substring(value, 145, 5)), "EFT-REC-DATE-LAST-UPDT", trim(substring(value, 150, 6)), "info", named_struct( "title", "provider", "version", "0.0.1", "baseUri", "http://scdhhs.gov/provider/" ) ) as instance, concat("/provider/eftprov/",md5(value), "/", uuid(), ".json") as fullurl from ( select value , array_repeat('a', 0) as EMPTY_ARRAY from fixedLengthInput ) '''RETURN sqlcmd
```

MMIS Provider SQL

```
1 get_mmis_provider_sql_query(): sqlcmd = u''' select named_struct( "PROV-NUMBER", trim(substring(value, 1, 6)), "PROCESS-IND", trim(substring(value, 7, 1)), "PROV-GROUP-IND", trim(substring(value, 8, 1)), "PROV-DIALYSIS-IND", trim(substring(value, 9, 1)), "PROV-RATE-IND", trim(substring(value, 10, 1)), "SOUND-SORT-KEY", named_struct( "PROV-SOUNDX-KEY", trim(substring(value, 11, 5)), "PROV-SORT-KEY", trim(substring(value, 16, 16)) ), "PROVIDER-NAME", trim(substring(value, 32, 26)), "PROV-LOCATION", named_struct( "PROV-CAREOF", trim(substring(value, 58, 26)), "PROV-STREET", trim(substring(value, 84, 26)), "PROV-CITY", trim(substring(value, 110, 20)), "PROV-STATE", trim(substring(value, 130, 2)), "PROV-ZIP", trim(substring(value, 132, 9)), "PROV-TEL-NO", trim(substring(value, 141, 10)) ), "PAYTO-LOCATION", named_struct( "PAYTO-CAREOF", trim(substring(value, 151, 26)), "PAYTO-STREET", trim(substring(value, 177, 26)), "PAYTO-CITY", trim(substring(value, 203, 20)), "PAYTO-STATE", trim(substring(value, 223, 2)) ), "PAYTO-ZIP", trim(substring(value, 225, 9)) ), "TPL-LOCATION", named_struct( "TPL-CAREOF", trim(substring(value, 234, 26)), "TPL-STREET", trim(substring(value, 260, 26)), "TPL-CITY", trim(substring(value, 286, 20)), "TPL-STATE", trim(substring(value, 306, 2)), "TPL-ZIP", trim(substring(value, 308, 9)) ), "PHYSICAL-LOCATION", named_struct( "PL-STREET", trim(substring(value, 317, 26)), "PL-CITY", trim(substring(value, 343, 20)), "PL-STATE", trim(substring(value, 363, 2)), "PL-ZIP", trim(substring(value, 365, 9)) ), "PROVIDER-TYPE", trim(substring(value, 374, 2)), "PROV-SERVICE-DATA", transform(split(trim(TRAILING "~" FROM regexp_replace(trim(substring(value, 376, (18 * 6))), "{.18}","$1~")), '~'), value -> named_struct( "PROV-CAT-SERV", trim(substring(value, 1, 2)), "PROV-BEG-SERV-DT", named_struct( "PROV-BEG-CN", trim(substring(value, 3, 2)), "PROV-BEG-SERV", trim(substring(value, 5, 6)) ), "PROV-END-SERV-DT", named_struct( "PROV-END-CN", trim(substring(value, 11, 2)), "PROV-END-SERV", trim(substring(value, 13, 6)) ) ) ), "TYPE-PRACTICE", trim(substring(value, 484, 2)), "PROV-TYPE-AFFIL", trim(substring(value, 486, 2)), "PROV-COUNTY", trim(substring(value, 488, 2)), "PROV-EMP-ID",
```

```

trim(substring(value, 490, 9)) , "PROV-SOC-SEC-NO", trim(substring(value, 499, 9)) , "PROV-YR-END-DATE",
named_struct( "PROV-YR-END-MO", trim(substring(value, 508, 2)) , "PROV-YR-END-DA", trim(substring(value,
510, 2)) ) , "PROV-LICENSE-BRD", trim(substring(value, 512, 2)) , "PROV-LICENSE-DATE", named_struct( "PROV-
LIC-YR", trim(substring(value, 514, 2)) , "PROV-LIC-MO", trim(substring(value, 516, 2)) , "PROV-LIC-DA",
trim(substring(value, 518, 2)) ) , "PROV-SPECIALTY-DATA", transform(split(trim(TRAILING "~" FROM
regexp_replace(trim(substring(value, 520, (8 * 3))), "(. {8})", "$1~")), '~'), value -> named_struct( "PROV-
SPECIALTY", trim(substring(value, 1, 2)) , "PROV-SPC-CRT-DT", trim(substring(value, 3, 6)) ) ) , "PROV-LST-
PAY-DATE", named_struct( "PROV-LST-PAY-YR", trim(substring(value, 544, 2)) , "PROV-LST-PAY-MO",
trim(substring(value, 546, 2)) , "PROV-LST-PAY-DA", trim(substring(value, 548, 2)) ) , "PROV-EXP-IND",
trim(substring(value, 550, 1)) , "PROV-DEBIT-INFO", named_struct( "PROV-DEBIT-BAL", translate(concat(
if(instr('}jklmnopqr', substring(value, 551 + 11 - 1, 1)) > 0, '-', '')) , if(trim(LEADING '0' FROM
substring(value, 551, if(9 > -1, 9, 11))) = '0', '')) , trim(LEADING '0' FROM substring(value, 551, if(9
> -1, 9, 11))) , if(9 > -1, '.', '')) , if(9 > -1, substring(value, 551 + 9, 11 - 9), '')) ) ,
'{ABCDEFGH}ijklmnopqr', '01234567890123456789') , "PROV-DEBIT-BAL-DATE", named_struct( "PROV-DB-YR",
trim(substring(value, 562, 2)) , "PROV-DB-MO", trim(substring(value, 564, 2)) , "PROV-DB-DA",
trim(substring(value, 566, 2)) ) ) , "PROV-DEBIT-SPACE", named_struct( "PROV-ELECTRIC-IND",
trim(substring(value, 551, 1)) , "PROV-AVAIL-SPACE", trim(substring(value, 552, 16)) ) ) , "PROV-APPL-DATE",
named_struct( "PROV-APPL-YR", trim(substring(value, 568, 2)) , "PROV-APPL-MO", trim(substring(value, 570,
2)) , "PROV-APPL-DA", trim(substring(value, 572, 2)) ) , "PROV-OUT-STATE-IND", trim(substring(value, 574,
1)) , "PROV-BNDD", trim(substring(value, 575, 9)) , "PROV-CHAIN-NAME", trim(substring(value, 584, 15)) ,
"PROV-BED-INFO", transform(split(trim(TRAILING "~" FROM regexp_replace(trim(substring(value, 599, (5 *
3))), "(. {5})", "$1~")), '~'), value -> named_struct( "PROV-BED-TYPE", trim(substring(value, 1, 1)) , "PROV-
NUM-BEDS", trim(substring(value, 2, 4)) , "PROV-NUM-BEDS-RED", int(translate(concat( if(instr('}jklmnopqr',
substring(value, 2 + 4 - 1, 1)) > 0, '-', '')) , if(trim(LEADING '0' FROM substring(value, 2, if(-1 > -1, -1,
4))) = '0', '')) , trim(LEADING '0' FROM substring(value, 2, if(-1 > -1, -1, 4))) , if(-1 > -1, '.', ''))
, if(-1 > -1, substring(value, 2 + -1, 4 - -1), '')) , '{ABCDEFGH}ijklmnopqr', '01234567890123456789') ) ) ,
"PROV-CNTRL-FACIL", trim(substring(value, 614, 3)) , "PROV-ENROLLMENT", transform(split(trim(TRAILING "~"
FROM regexp_replace(trim(substring(value, 617, (9 * 5))), "(. {9})", "$1~")), '~'),
2 value -> named_struct( "PROV-ENROL-STAT", trim(substring(value, 1, 1)) , "DECODE-1",
trim(substring(value, 1, 1)) , "PROV-ENROL-DT", named_struct( "PROV-ENROL-CN", trim(substring(value, 2, 2))
, "PROV-ENROL-DATE", trim(substring(value, 4, 6)) ) ) ) , "PROV-NUM-PHYSICIANS", trim(substring(value, 662,
3)) , "PROV-COMPOSITE-IND", trim(substring(value, 665, 1)) , "PROV-DATE-CREATED", trim(substring(value, 666,
6)) , "PROV-ALERT-CANCEL-DT", trim(substring(value, 672, 6)) , "PROV-NUM-HMO-MEMBS", trim(substring(value,
678, 5)) , "PROV-DEGREE", trim(substring(value, 683, 3)) , "PROV-CERT-BEGIN", trim(substring(value, 686, 6))
, "PROV-CERT-END", trim(substring(value, 692, 6)) , "DATE-OF-LAST-UPDATE", named_struct( "DLU-YR",
trim(substring(value, 698, 2)) , "DLU-MO", trim(substring(value, 700, 2)) , "DLU-DA", trim(substring(value,
702, 2)) ) , "ID-OF-LAST-UPDATE", trim(substring(value, 704, 5)) , "PROV-ERROR-DATA", named_struct( "PROV-
ERR-INFO", transform(split(trim(TRAILING "~" FROM regexp_replace(trim(substring(value, 709, (3 * 10))), "(.
{3})", "$1~")), '~'), value -> named_struct( "PROV-ERR-NO", trim(substring(value, 1, 3)) ) ) ) , "NCPDP-NO",
trim(substring(value, 739, 7)) , "NCPDP-DATE", named_struct( "NCPDP-CN", trim(substring(value, 746, 2)) ,
"NCPDP-YMD", named_struct( "NCPDP-YR", trim(substring(value, 748, 2)) , "NCPDP-MO", trim(substring(value,
750, 2)) , "NCPDP-DA", trim(substring(value, 752, 2)) ) ) , "NAT-ID-PROV-TYPE", trim(substring(value, 754,
1)) , "NAT-PROV-ID", trim(substring(value, 755, 10)) , "FILLER", trim(substring(value, 765, 9)) , "PROV-
RATE-INFO", named_struct( "PROV-CHRG-MODE", trim(substring(value, 774, 1)) , "PREG-IND",
trim(substring(value, 775, 1)) , "UNDER21-IND", trim(substring(value, 776, 1)) , "PAYMENT-INDICATOR",
trim(substring(value, 777, 1)) , "PROV-1099-IND", trim(substring(value, 778, 1)) , "MGC-REC-IND",
trim(substring(value, 779, 1)) , "ACCESS-IND", trim(substring(value, 780, 3)) , "FILLER",
trim(substring(value, 783, 18)) , "PROV-NO-SERVICES", int(translate(concat( if(instr('}jklmnopqr',
substring(value, 801 + 2 - 1, 1)) > 0, '-', '')) , if(trim(LEADING '0' FROM substring(value, 801, if(-1 > -1,
-1, 2))) = '0', '')) , trim(LEADING '0' FROM substring(value, 801, if(-1 > -1, -1, 2))) , if(-1 > -1,
',', '')) , if(-1 > -1, substring(value, 801 + -1, 2 - -1), '')) , '{ABCDEFGH}ijklmnopqr',
'01234567890123456789') ) , "PROV-SERVICE-SETS", transform(split(trim(TRAILING "~" FROM
regexp_replace(trim(substring(value, 803, (186 * 3))), "(. {186})", "$1~")), '~'), value -> named_struct(
"PROV-SERVICE", trim(substring(value, 1, 6)) , "PROV-RATE-SETS", transform(split(trim(TRAILING "~" FROM
regexp_replace(trim(substring(value, 7, (15 * 12))), "(. {15})", "$1~")), '~'), value -> named_struct( "PROV-
RATE", translate(concat( if(instr('}jklmnopqr', substring(value, 1 + 7 - 1, 1)) > 0, '-', '')) ,
if(trim(LEADING '0' FROM substring(value, 1, if(5 > -1, 5, 7))) = '0', '')) , trim(LEADING '0' FROM
substring(value, 1, if(5 > -1, 5, 7))) , if(5 > -1, '.', '')) , if(5 > -1, substring(value, 1 + 5, 7 - 5),
'')) , '{ABCDEFGH}ijklmnopqr', '01234567890123456789') , "PROV-RATE-DT", named_struct( "PROV-RATE-CN",
trim(substring(value, 8, 2)) , "PROV-RATE-EFF", trim(substring(value, 10, 6)) ) ) ) ) , "info",
named_struct( "title", "provider", "version", "0.0.1", "baseUrl", "http://scdhhs.gov/provider/" ) ) as
instance, concat("/provider/mmispv/", md5(value), "/", uuid(), ".json") as fullurl from ( select value ,
array_repeat('a', 0) as EMPTY_ARRAY from fixedLengthInput ) '''RETURN sqlcmd
3
4

```

NPI Crosswalk SQL

```

1 get_npi_crosswalk_sql_query(): sqlcmd = u''' select named_struct( "NPI-NAT-PROV-ID", trim(substring(value,
1, 10)) , "NPI-NAT-DATE", named_struct( "NPI-NAT-CN", trim(substring(value, 11, 2)) , "NPI-NAT-YMD",
named_struct( "NPI-NAT-YR", trim(substring(value, 13, 2)) , "NPI-NAT-MO", trim(substring(value, 15, 2)) ,
"NPI-NAT-DA", trim(substring(value, 17, 2)) ) ) , "NPI-PROV-GROUP-IND", trim(substring(value, 19, 1)) ,
"NPI-NAT-SUBMIT-SOURCE", if(trim(substring(value, 20, (1 * 3))) = '', EMPTY_ARRAY,
transform(split(trim(TRAILING "~" FROM regexp_replace(trim(substring(value, 20, (1 * 3))), "(. {1})", "$1~")),
'~'), value -> trim(value))) , "NPI-SUB-RSN-INFO", transform(split(trim(TRAILING "~" FROM
regexp_replace(trim(substring(value, 23, (48 * 3))), "(. {48})", "$1~")), '~'), value -> named_struct( "NPI-

```



```

SUBMIT-REASON", trim(substring(value, 1, 1)) , "NPI-SUBMIT-REASON-NPI", trim(substring(value, 2, 10)) ,
"NPI-SUBMIT-DEACT-RSN", trim(substring(value, 12, 1)) , "NPI-TERM-DATE", named_struct( "NPI-TERM-CC",
trim(substring(value, 13, 2)) , "NPI-TERM-YMD", named_struct( "NPI-TERM-YY", trim(substring(value, 15, 2)) ,
"NPI-TERM-MM", trim(substring(value, 17, 2)) , "NPI-TERM-DD", trim(substring(value, 19, 2)) ) ) , "NPI-
REACT-DATE", named_struct( "NPI-REACT-CN", trim(substring(value, 21, 2)) , "NPI-REACT-YMD", named_struct(
"NPI-REACT-YY", trim(substring(value, 23, 2)) , "NPI-REACT-MM", trim(substring(value, 25, 2)) , "NPI-REACT-
DD", trim(substring(value, 27, 2)) ) ) , "NPI-REPLACE-NPI", trim(substring(value, 29, 10)) , "NPI-PREVIOUS-
NPI", trim(substring(value, 39, 10)) ) ) , "NPI-DATE-OF-LAST-UPDATE", named_struct( "NPI-DLU-YR",
trim(substring(value, 167, 2)) , "NPI-DLU-MO", trim(substring(value, 169, 2)) , "NPI-DLU-DA",
trim(substring(value, 171, 2)) ) , "NPI-ID-OF-LAST-UPDATE", trim(substring(value, 173, 5)) , "NPI-NAT-ID-
VERIFY-IND", trim(substring(value, 178, 1)) , "NPI-NAT-IND-INFO", named_struct( "NPI-NAT-PRIM-FNAME",
trim(substring(value, 179, 20)) , "NPI-NAT-PRIM-MNAME", trim(substring(value, 199, 20)) , "NPI-NAT-PRIM-
LNAME", trim(substring(value, 219, 35)) , "NPI-NAT-OTHR-FNAME", trim(substring(value, 254, 20)) , "NPI-NAT-
OTHR-MNAME", trim(substring(value, 274, 20)) , "NPI-NAT-OTHR-LNAME", trim(substring(value, 294, 35)) , "NPI-
NAT-NAME-TYPE", trim(substring(value, 329, 1)) , "NPI-NAT-IRS-ID", trim(substring(value, 330, 9)) , "NPI-
PROV-SOC-SEC-NO", trim(substring(value, 339, 9)) ) , "NPI-NAT-ORG-INFO", named_struct( "NPI-NAT-ORG-NAME",
trim(substring(value, 348, 70)) , "NPI-NAT-ORG-OTH-NAME", trim(substring(value, 418, 70)) , "NPI-NAT-TYPE-
OTH-NAME", trim(substring(value, 488, 1)) , "NPI-PROV-EMP-ID", trim(substring(value, 489, 9)) ) , "NPI-NAT-
ADDR-INFO", named_struct( "NPI-NAT-MAIL-ADDR-L1", trim(substring(value, 498, 55)) , "NPI-NAT-MAIL-ADDR-L2",
trim(substring(value, 553, 55)) , "NPI-NAT-MAIL-CITY", trim(substring(value, 608, 40)) , "NPI-NAT-MAIL-
STATE", trim(substring(value, 648, 2)) , "NPI-NAT-MAIL-ZIP", named_struct( "NPI-NAT-MAIL-ZIP-5",
trim(substring(value, 650, 5)) , "NPI-NAT-MAIL-ZIP-4", trim(substring(value, 655, 4)) ) , "NPI-NAT-MAIL-
PHONE", named_struct( "NPI-NAT-MAIL-AREA", trim(substring(value, 659, 3)) , "NPI-NAT-MAIL-PRE",
trim(substring(value, 662, 3)) , "NPI-NAT-MAIL-LAST-4", trim(substring(value, 665, 4)) ) , "NPI-NAT-MAIL-
FAX", named_struct( "NPI-NAT-MAIL-FXAREA", trim(substring(value, 669, 3)) , "NPI-NAT-MAIL-FXPRES",
trim(substring(value, 672, 3)) , "NPI-NAT-MAIL-FX-L4", trim(substring(value, 675, 4)) ) , "NPI-NAT-PRAC-
ADDR-L1", trim(substring(value, 679, 55)) , "NPI-NAT-PRAC-ADDR-L2", trim(substring(value, 734, 55)) , "NPI-
NAT-PRAC-CITY", trim(substring(value, 789, 40)) , "NPI-NAT-PRAC-STATE", trim(substring(value, 829, 2)) ,
"NPI-NAT-PRAC-ZIP", named_struct( "NPI-NAT-PRAC-ZIP-5", trim(substring(value, 831, 5)) , "NPI-NAT-PRAC-ZIP-
4", trim(substring(value, 836, 4)) ) , "NPI-NAT-PRAC-PHONE", named_struct( "NPI-NAT-PRAC-AREA",
trim(substring(value, 840, 3)) , "NPI-NAT-PRAC-PRE", trim(substring(value, 843, 3)) , "NPI-NAT-PRAC-LAST-4",
trim(substring(value, 846, 4)) ) , "NPI-NAT-PRAC-FAX", named_struct( "NPI-NAT-PRAC-FXAREA",
trim(substring(value, 850, 3)) , "NPI-NAT-PRAC-FXPRES", trim(substring(value, 853, 3)) , "NPI-NAT-PRAC-FX-
L4", trim(substring(value, 856, 4)) ) ) , "NPI-NAT-UPIN", trim(substring(value, 860, 20)) , "NPI-NAT-
MEDICARE", trim(substring(value, 880, 20)) , "NPI-NAT-MDCR-STATE", trim(substring(value, 900, 2)) , "NPI-
NAT-MEDICAID", trim(substring(value, 902, 20)) , "NPI-NAT-MDCD-STATE", trim(substring(value, 922, 2)) ,
"NPI-NAT-OTHER-ID", if(trim(substring(value, 924, (20 * 2))) = '' , EMPTY_ARRAY,
transform(split(trim(TRAILING "~" FROM regexp_replace(trim(substring(value, 924, (20 * 2)))), "(.
{20})", "$1~")) , '~') , value -> trim(value))) , "NPI-NAT-OTHER-STATE", if(trim(substring(value, 964, (2 *
2))) = '' , EMPTY_ARRAY, transform(split(trim(TRAILING "~" FROM regexp_replace(trim(substring(value, 964, (2
* 2))), "(.{2})", "$1~")) , '~') , value -> trim(value))) , "NPI-NAT-OTH-ISSUER", if(trim(substring(value, 968,
(25 * 2))) = '' , EMPTY_ARRAY, transform(split(trim(TRAILING "~" FROM regexp_replace(trim(substring(value,
968, (25 * 2))), "(.{25})", "$1~")) , '~') , value -> trim(value))) ,
2 ' , value -> trim(value))) , "NPI-NAT-CONT-FNAME", trim(substring(value, 1018, 20)) , "NPI-NAT-CONT-MNAME",
trim(substring(value, 1038, 20)) , "NPI-NAT-CONT-LNAME", trim(substring(value, 1058, 35)) , "NPI-NAT-CONT-
TITLE", trim(substring(value, 1093, 35)) , "NPI-NAT-CONT-PHONE", named_struct( "NPI-NAT-CONT-AREA",
trim(substring(value, 1128, 3)) , "NPI-NAT-CONT-PRE", trim(substring(value, 1131, 3)) , "NPI-NAT-CONT-LAST-
4", trim(substring(value, 1134, 4)) ) , "NPI-PROV-NUMBER", trim(substring(value, 1138, 6)) , "NPI-PAYMENT-
INDICATOR", trim(substring(value, 1144, 1)) , "NPI-PROV-CNTRL-FACIL", trim(substring(value, 1145, 3)) ,
"NPI-PROV-ELECTRIC-IND", trim(substring(value, 1148, 1)) , "NPI-PAYTO-LOCATION", named_struct( "NPI-PAYTO-
CAREOF", trim(substring(value, 1149, 26)) , "NPI-PAYTO-STREET", trim(substring(value, 1175, 26)) , "NPI-
PAYTO-CITY", trim(substring(value, 1201, 20)) , "NPI-PAYTO-STATE", trim(substring(value, 1221, 2)) , "NPI-
PAYTO-ZIP", trim(substring(value, 1223, 9)) ) , "info", named_struct( "title", "provider", "version",
"0.0.1", "baseUri", "http://scdhhs.gov/provider/" ) ) as instance, concat("/provider/npjunk/", md5(value),
"/", uuid(), ".json") as fullurl from ( select value , array_repeat('a', 0) as EMPTY_ARRAY from
fixedLengthInput ) '''RETURN sqlcmd

```

NPI Junction SQL

```

1 get_npi_junction_sql_query(): sqlcmd = u''' select named_struct( "NPJ-NAT-PROV-ID", trim(substring(value, 1,
10)) , "NPJ-PROV-NUMBER", trim(substring(value, 11, 6)) , "NPJ-TAXONOMY-NUM", trim(substring(value, 17, 10))
, "NPJ-ZIP-5", trim(substring(value, 27, 5)) , "NPJ-ZIP-4", trim(substring(value, 32, 4)) , "NPJ-DOC-TYPE",
trim(substring(value, 36, 1)) , "NPJ-CDE-SRC", trim(substring(value, 37, 2)) , "NPJ-PROVIDER-TYPE",
trim(substring(value, 39, 2)) , "NPJ-PROV-SPECIALTY", trim(substring(value, 41, 2)) , "NPJ-INDICATOR",
trim(substring(value, 43, 1)) , "NPJ-NAT-DATE", named_struct( "NPJ-NAT-CN", trim(substring(value, 44, 2)) ,
"NPI-NAT-YMD", named_struct( "NPJ-NAT-YR", trim(substring(value, 46, 2)) , "NPJ-NAT-MO",
trim(substring(value, 48, 2)) , "NPJ-NAT-DA", trim(substring(value, 50, 2)) ) ) , "NPJ-FLAG-1",
trim(substring(value, 52, 1)) , "NPJ-FLAG-2", trim(substring(value, 53, 1)) , "NPJ-MISC-IND-1",
trim(substring(value, 54, 1)) , "info", named_struct( "title", "provider", "version", "0.0.1", "baseUri",
"http://scdhhs.gov/provider/" ) ) as instance, concat("/provider/npjunk/", md5(value), "/", uuid(), ".json")
as fullurl from ( select value , array_repeat('a', 0) as EMPTY_ARRAY from fixedLengthInput ) '''RETURN
sqlcmd

```

Auxiliary SQL

```

1 get_prov_auxiliary_sql_query(): sqlcmd = u''' select named_struct( "PAX-PROV-NUM", trim(substring(value, 1, 6)), "PAX-NPI-NUM", trim(substring(value, 7, 10)), "PAX-MEDICARE-INFO", named_struct( "PAX-MEDICARE-NO", trim(substring(value, 17, 12)) , "PAX-MED-END", trim(substring(value, 29, 2)) ) , "PAX-NEW-LIC-INFO", named_struct( "PAX-PROV-LIC-NUM", trim(substring(value, 17, 6)) , "PAX-PROVIDER-TYPE", trim(substring(value, 23, 2)) , "PAX-LIC-BOARD", trim(substring(value, 25, 2)) , "PAX-PRACTICE-SPECIALTY", trim(substring(value, 27, 2)) , "PAX-LIC-END-2", trim(substring(value, 29, 2)) ) , "PAX-FH-LIC-INFO", named_struct( "PAX-FH-LICENSE-NUMBER", trim(substring(value, 17, 10)) , "PAX-FH-END", trim(substring(value, 27, 4)) ) , "PAX-NUM-INDIC-INFO", named_struct( "PAX-NUM-INDIC", trim(substring(value, 31, 1)) , "PAX-BC-IND", trim(substring(value, 32, 1)) ) , "info", named_struct( "title", "provider", "version", "0.0.1", "baseUri", "http://scdhhs.gov/provider/" ) ) as instance, concat("/provider/provaux/",md5(value), "/", uuid(), ".json") as fullurl from ( select value , array_repeat('a', 0) as EMPTY_ARRAY from fixedLengthInput )

```

Affiliation SQL

```

1 get_provider_affiliation_sql_query(): sqlcmd = u''' select named_struct( "PAE-GROUP-PROV-CODE", trim(substring(value, 1, 6)) , "PAE-FILL1", trim(substring(value, 7, 1)) , "PAE-GROUP-NPI-CODE", trim(substring(value, 8, 10)) , "PAE-FILL2", trim(substring(value, 18, 1)) , "PAE-IND-PROV-CODE", trim(substring(value, 19, 6)) , "PAE-FILL3", trim(substring(value, 25, 1)) , "PAE-IND-NPI-CODE", trim(substring(value, 26, 10)) , "PAE-FILL4", trim(substring(value, 37, 1)) , "info", named_struct( "title", "provider", "version", "0.0.1", "baseUri", "http://scdhhs.gov/provider/" ) ) as instance, concat("/provider/provaux/",md5(value), "/", uuid(), ".json") as fullurl from ( select value , array_repeat('a', 0) as EMPTY_ARRAY from fixedLengthInput ) '''RETURN sqlcmd

```

Primary Tin SQL

```

1 get_prim_tin_sql_query(): sqlcmd = u''' select named_struct( "TIN-TIN", trim(substring(value, 1, 9)) , "TIN-MMIS-NUMBER", named_struct( "TIN-PROVIDER-LEGACY", trim(substring(value, 10, 6)) , "TIN-PROVIDER-NPI", trim(substring(value, 16, 10)) ) , "TIN-EFFECT-BEGIN-DATE", trim(substring(value, 26, 8)) , "TIN-TERMINATE-DATE", trim(substring(value, 34, 8)) , "info", named_struct( "title", "provider", "version", "0.0.1", "baseUri", "http://scdhhs.gov/provider/" ) ) as instance, concat("/provider/primtin/",md5(value), "/", uuid(), ".json") as fullurl from ( select value , array_repeat('a', 0) as EMPTY_ARRAY from fixedLengthInput ) '''RETURN sqlcmd

```

Scies Vendor SQL

```

1 get_sceis_vendor_sql_query(): sqlcmd = u''' select named_struct( "VENDOR-S-V-KEY", named_struct( "VENDOR-TIN", trim(substring(value, 1, 9)) , "VENDOR-PRIMARY-TIN-IND", trim(substring(value, 10, 2)) ) , "VENDOR-SCEIS-NUMBER", trim(substring(value, 12, 10)) , "VENDOR-SCEIS-BLOCK-IND", trim(substring(value, 22, 2)) , "VENDOR-EFFECT-BEGIN-DATE", trim(substring(value, 24, 8)) , "VENDOR-TERMINATE-DATE", trim(substring(value, 32, 8)) , "info", named_struct( "title", "provider", "version", "0.0.1", "baseUri", "http://scdhhs.gov/provider/" ) ) as instance, concat("/provider/sceisvn/",md5(value), "/", uuid(), ".json") as fullurl from ( select value , array_repeat('a', 0) as EMPTY_ARRAY from fixedLengthInput ) '''RETURN sqlcmd

```

Provider canonical structure

```

1 {
2   "_id": "1003897125",
3   "envelope": {
4     "headers": {
5       "metadata": {
6         "applicationType": "MES",
7         "batchId": "547418fcf263437a90bbac870497cfcf",
8         "createdBy": "admin",
9         "createdTime": "2024-09-17T09:40:25.774018-0400",
10        "hashValue": "e24390de31c059f5b1e46350ef81da1",
11        "jobName": "blue-provider-mmis-typical-in",
12        "jobType": "6blue",
13        "lastUpdatedBy": "admin",
14        "lastUpdatedTime": "2024-09-17T09:40:25.774077-0400",
15        "npi": "1003897125",
16        "processName": "Ingestion",
17        "version": "1.0"
18      }
19    },
20    "instance": {
21      "provider": {
22        "providerAssociate": [],
23        "providerCertification": [
24          {
25            "deaSchedule": "",

```



```

26 "effectiveDate": "1999-03-19",
27 "expirationDate": "2000-06-19",
28 "issuingAuthority": "",
29 "number": "",
30 "state": "",
31 "status": "MMIS-Provider",
32 "type": "",
33 "verificationIndicator": ""
34 },
35 {
36 "deaSchedule": "",
37 "effectiveDate": "2012-02-01",
38 "expirationDate": "2013-01-31",
39 "issuingAuthority": "",
40 "number": "",
41 "state": "",
42 "status": "MMIS-Provider",
43 "type": "",
44 "verificationIndicator": ""
45 },
46 {
47 "deaSchedule": "",
48 "effectiveDate": "2001-07-25",
49 "expirationDate": "2022-01-24",
50 "issuingAuthority": "CLIA",
51 "number": "42D0903567",
52 "state": "",
53 "status": "MMIS-Clia",
54 "type": "Certificate of Waiver (CoW)",
55 "verificationIndicator": "A"
56 }
57 ],
58 "providerEnrollment": [
59 {
60 "applicationDate": "",
61 "confirmationID": "",
62 "effectiveDate": "1998-01-30",
63 "enrollmentDate": "",
64 "expirationDate": "",
65 "id": "",
66 "identifiers": {
67 "MMIS": "0737NF",
68 "NPI": "1003897125"
69 },
70 "isFeeCollected": "True",
71 "isOutOfState": "False",
72 "isWaiver": "N",
73 "providerLocation": {
74 "PCMHNumber": "",
75 "address": {
76 "addresstype": [
77 "PRIMARY",
78 "PRACTICE"
79 ],
80 "city": "MANNING",
81 "county": "Clarendon",
82 "countyCode": "14",
83 "line1": "5583 SUMMERTON HWY",
84 "line2": "",
85 "line3": "",
86 "state": "SC",
87 "zip": "291025217"
88 },
89 "contactPerson": [],
90 "email": "",
91 "facilities": {
92 "bedsInfo": [
93 {
94 "count": "0064",
95 "isRequired": "",
96 "type": "SNF"
97 }
98 ],
99 "facilityFeature": []
100 },
101 "faxNumber": "",

```

```
102 "fiscaldate": "",
103 "hours": [],
104 "id": "",
105 "isAcceptingNewPatients": "",
106 "isSiblingAccepted": "",
107 "languages": [],
108 "name": "WINDSOR MANOR NURSING FACILITY",
109 "phoneNumber": {
110   "ext": "",
111   "number": "8034782323"
112 },
113 },
114 "providerSpecialty": {
115   "description": "",
116   "docTypeCode": "G",
117   "docTypeDescription": "NPJ-NURSE-HOME-INV",
118   "effectiveDate": "1998-01-30",
119   "indicatorCode": "7",
120   "indicatorCodeDescription": "ACT DO NOT PAY T 18",
121   "isprimary": "Y",
122   "providerServiceCategory": [
123     {
124       "categoryCode": "11",
125       "categoryDescription": "SKILLED NURSING FAC",
126       "effectiveDate": "",
127       "expirationDate": ""
128     }
129   ],
130   "providerSubSpeciality": "",
131   "providerType": "00",
132   "providerTypedescription": "NURSING HOME",
133   "specialityCode": "",
134   "taxonomy": "314000000X",
135   "taxonomyDescription": "",
136   "terminationDate": "",
137   "zipCode": "29102"
138 },
139 "status": "ENROLLED",
140 "type": "FA",
141 "waiverReason": ""
142 },
143 {
144   "applicationDate": "",
145   "confirmationID": "",
146   "effectiveDate": "2012-07-01",
147   "enrollmentDate": "",
148   "expirationDate": "",
149   "id": "",
150   "identifiers": {
151     "MMIS": "C737NF",
152     "NPI": "1003897125"
153   },
154   "isFeeCollected": "True",
155   "isOutOfState": "False",
156   "isWaiver": "N",
157   "providerLocation": {
158     "PCMHNumber": "",
159     "address": {
160       "addresstype": [
161         "PRIMARY",
162         "PRACTICE"
163       ],
164       "city": "MANNING",
165       "county": "Clarendon",
166       "countyCode": "14",
167       "line1": "5583 SUMMERTON HWY",
168       "line2": "",
169       "line3": "",
170       "state": "SC",
171       "zip": "291025217"
172     },
173     "contactPerson": [],
174     "email": "",
175     "facilities": {
176       "bedsInfo": [
177         {
```

```
178 "count": "0064",
179 "isRequired": "",
180 "type": "SNF"
181 }
182 ],
183 "facilityFeature": []
184 },
185 "faxNumber": "",
186 "fiscaldate": "",
187 "hours": [],
188 "id": "",
189 "isAcceptingNewPatients": "",
190 "isSiblingAccepted": "",
191 "languages": [],
192 "name": "WINDSOR MANOR NURSING FACILITY",
193 "phoneNumber": {
194   "ext": "",
195   "number": "8034782323"
196 }
197 },
198 "providerSpecialty": {
199   "description": "",
200   "docTypeCode": "G",
201   "docTypeDescription": "NPJ-NURSE-HOME-INV",
202   "effectiveDate": "1998-01-30",
203   "indicatorCode": "7",
204   "indicatorCodeDescription": "ACT DO NOT PAY T 18",
205   "isprimary": "Y",
206   "providerServiceCategory": [
207     {
208       "categoryCode": "11",
209       "categoryDescription": "SKILLED NURSING FAC",
210       "effectiveDate": "",
211       "expirationDate": ""
212     }
213   ],
214   "providerSubSpecialty": "",
215   "providerType": "00",
216   "providerTypedescription": "NURSING HOME",
217   "specialtyCode": "",
218   "taxonomy": "314000000X",
219   "taxonomyDescription": "",
220   "terminationDate": "",
221   "zipCode": "291025217"
222 },
223 "status": "ENROLLED",
224 "type": "FA",
225 "waiverReason": ""
226 }
227 ],
228 "providerLicense": [
229   {
230     "category": "",
231     "effectiveDate": "1999-02-01",
232     "expirationDate": "",
233     "issuingauthority": "40",
234     "number": "",
235     "state": "",
236     "status": "MMIS-Provider",
237     "type": ""
238   },
239   {
240     "category": "",
241     "effectiveDate": "2012-02-01",
242     "expirationDate": "",
243     "issuingauthority": "40",
244     "number": "",
245     "state": "",
246     "status": "MMIS-Provider",
247     "type": ""
248   },
249   {
250     "category": "",
251     "effectiveDate": "",
252     "expirationDate": "",
253     "issuingauthority": "40",
```

```
254 "number": "737",
255 "state": "",
256 "status": "PROV-Auxiliary",
257 "type": "00"
258 },
259 {
260 "category": "",
261 "effectiveDate": "",
262 "expirationDate": "",
263 "issuingauthority": "40",
264 "number": "NCF737",
265 "state": "",
266 "status": "PROV-Auxiliary",
267 "type": "00"
268 },
269 ],
270 "providerLocation": [
271 {
272 "PCMHNumber": "",
273 "address": {
274 "addresstype": [
275 "MAILING"
276 ],
277 "city": "MANNING",
278 "controlFacility": "001",
279 "county": "Clarendon",
280 "countyCode": "14",
281 "line1": "5583 SUMMERTON HWY",
282 "line2": "",
283 "line3": "",
284 "state": "SC",
285 "zip": "291025217"
286 },
287 "contactPerson": [],
288 "email": "",
289 "facilities": {
290 "bedsInfo": [],
291 "facilityFeature": []
292 },
293 "faxNumber": "",
294 "fiscaldate": "",
295 "hours": [],
296 "id": "",
297 "identifiers": {
298 "MMIS": "0737NF",
299 "NPI": "1003897125"
300 },
301 "isAcceptingNewPatients": "",
302 "isSiblingAccepted": "",
303 "languages": [],
304 "name": "",
305 "phoneNumber": {
306 "ext": "",
307 "number": "8034782323"
308 }
309 },
310 {
311 "PCMHNumber": "",
312 "address": {
313 "addresstype": [
314 "BILLING"
315 ],
316 "city": "MANNING",
317 "county": "Clarendon",
318 "countyCode": "14",
319 "line1": "PO BOX 57",
320 "line2": "",
321 "line3": "",
322 "state": "SC",
323 "zip": "291020057"
324 },
325 "contactPerson": [],
326 "email": "",
327 "facilities": {
328 "bedsInfo": [],
329 "facilityFeature": []
```

```

330 },
331 "faxNumber": "",
332 "fiscaldate": "",
333 "hours": [],
334 "id": "",
335 "identifiers": {
336 "MMIS": "0737NF",
337 "NPI": "1003897125"
338 },
339 "isAcceptingNewPatients": "",
340 "isSiblingAccepted": "",
341 "languages": [],
342 "name": "CLARENDON HEALTH SYSTEM LO",
343 "phoneNumber": {
344 "ext": "",
345 "number": ""
346 }
347 },
348 {
349 "PCMHNumber": "",
350 "address": {
351 "addresstype": [
352 "PRIMARY",
353 "PRACTICE"
354 ],
355 "city": "MANNING",
356 "county": "Clarendon",
357 "countyCode": "14",
358 "line1": "5583 SUMMERTON HWY",
359 "line2": "",
360 "line3": "",
361 "state": "SC",
362 "zip": "291025217"
363 },
364 "contactPerson": [],
365 "email": "",
366 "facilities": {
367 "bedsInfo": [
368 {
369 "count": "0064",
370 "isRequired": "",
371 "type": "SNF"
372 }
373 ],
374 "facilityFeature": []
375 },
376 "faxNumber": "",
377 "fiscaldate": "",
378 "hours": [],
379 "id": "",
380 "identifiers": {
381 "MMIS": "0737NF",
382 "NPI": "1003897125"
383 },
384 "isAcceptingNewPatients": "",
385 "isSiblingAccepted": "",
386 "languages": [],
387 "name": "WINDSOR MANOR NURSING FACILITY",
388 "phoneNumber": {
389 "ext": "",
390 "number": "8034782323"
391 }
392 },
393 {
394 "PCMHNumber": "",
395 "address": {
396 "addresstype": [
397 "MAILING"
398 ],
399 "city": "MANNING",
400 "controlFacility": "001",
401 "county": "Clarendon",
402 "countyCode": "14",
403 "line1": "5583 SUMMERTON HWY",
404 "line2": "",
405 "line3": "",

```



```
406 "state": "SC",
407 "zip": "291025217"
408 },
409 "contactPerson": [],
410 "email": "",
411 "facilities": {
412 "bedsInfo": [],
413 "facilityFeature": []
414 },
415 "faxNumber": "",
416 "fiscaldate": "",
417 "hours": [],
418 "id": "",
419 "identifiers": {
420 "MMIS": "C737NF",
421 "NPI": "1003897125"
422 },
423 "isAcceptingNewPatients": "",
424 "isSiblingAccepted": "",
425 "languages": [],
426 "name": "",
427 "phoneNumber": {
428 "ext": "",
429 "number": "8034782323"
430 }
431 },
432 {
433 "PCMHNumber": "",
434 "address": {
435 "addresstype": [
436 "BILLING"
437 ],
438 "city": "MANNING",
439 "county": "Clarendon",
440 "countyCode": "14",
441 "line1": "PO BOX 57",
442 "line2": "",
443 "line3": "",
444 "state": "SC",
445 "zip": "291020057"
446 },
447 "contactPerson": [],
448 "email": "",
449 "facilities": {
450 "bedsInfo": [],
451 "facilityFeature": []
452 },
453 "faxNumber": "",
454 "fiscaldate": "",
455 "hours": [],
456 "id": "",
457 "identifiers": {
458 "MMIS": "C737NF",
459 "NPI": "1003897125"
460 },
461 "isAcceptingNewPatients": "",
462 "isSiblingAccepted": "",
463 "languages": [],
464 "name": "CLARENDON HEALTH SYSTEM LO",
465 "phoneNumber": {
466 "ext": "",
467 "number": ""
468 }
469 },
470 {
471 "PCMHNumber": "",
472 "address": {
473 "addresstype": [
474 "PRIMARY",
475 "PRACTICE"
476 ],
477 "city": "MANNING",
478 "county": "Clarendon",
479 "countyCode": "14",
480 "line1": "5583 SUMMERTON HWY",
481 "line2": "",
```

```
482 "line3": "",
483 "state": "SC",
484 "zip": "291025217"
485 },
486 "contactPerson": [],
487 "email": "",
488 "facilities": {
489 "bedsInfo": [
490 {
491 "count": "0064",
492 "isRequired": "",
493 "type": "SNF"
494 }
495 ],
496 "facilityFeature": []
497 },
498 "faxNumber": "",
499 "fiscaldate": "",
500 "hours": [],
501 "id": "",
502 "identifiers": {
503 "MMIS": "C737NF",
504 "NPI": "1003897125"
505 },
506 "isAcceptingNewPatients": "",
507 "isSiblingAccepted": "",
508 "languages": [],
509 "name": "WINDSOR MANOR NURSING FACILITY",
510 "phoneNumber": {
511 "ext": "",
512 "number": "8034782323"
513 }
514 },
515 ],
516 "providerPaymentMethod": [
517 {
518 "bankInfo": {
519 "accountHolder": "JOHNNIE GILLEY",
520 "accountNumber": "000000000000",
521 "accountType": "Checking",
522 "address": {
523 "city": "",
524 "line1": "",
525 "line2": "",
526 "state": "",
527 "zip": ""
528 },
529 "bankName": "",
530 "branchName": "",
531 "faxNumber": "",
532 "phoneNumber": {
533 "ext": "",
534 "number": "8034782323"
535 },
536 "routingNumber": "000000006"
537 },
538 "effectiveDate": "",
539 "expirationDate": "",
540 "paymentMethodType": "EFT"
541 }
542 ],
543 "providerSpeciality": [
544 {
545 "description": "",
546 "docTypeCode": "G",
547 "docTypeDescription": "NPJ-NURSE-HOME-INV",
548 "effectiveDate": "1998-01-30",
549 "identifiers": {
550 "MMIS": "0737NF",
551 "NPI": "1003897125"
552 },
553 "indicatorCode": "7",
554 "indicatorCodeDescription": "ACT DO NOT PAY T 18",
555 "isprimary": "Y",
556 "providerServiceCategory": [
557 {
```

```

558 "categoryCode": "11",
559 "categoryDescription": "SKILLED NURSING FAC",
560 "effectiveDate": "",
561 "expirationDate": ""
562 },
563 ],
564 "providerSubSpeciality": "",
565 "providerType": "00",
566 "providerTypedescription": "NURSING HOME",
567 "specialityCode": "",
568 "taxonomy": "314000000X",
569 "taxonomyDescription": "",
570 "terminationDate": "",
571 "zipCode": "29102"
572 },
573 {
574 "description": "",
575 "docTypeCode": "6",
576 "docTypeDescription": "NPJ-NURSE-HOME-INV",
577 "effectiveDate": "1998-01-30",
578 "identifiers": {
579 "MMIS": "C737NF",
580 "NPI": "1003897125"
581 },
582 "indicatorCode": "7",
583 "indicatorCodeDescription": "ACT DO NOT PAY T 18",
584 "isprimary": "Y",
585 "providerServiceCategory": [
586 {
587 "categoryCode": "11",
588 "categoryDescription": "SKILLED NURSING FAC",
589 "effectiveDate": "",
590 "expirationDate": ""
591 }
592 ],
593 "providerSubSpeciality": "",
594 "providerType": "00",
595 "providerTypedescription": "NURSING HOME",
596 "specialityCode": "",
597 "taxonomy": "314000000X",
598 "taxonomyDescription": "",
599 "terminationDate": "",
600 "zipCode": "291025217"
601 }
602 ],
603 "summaryOfSpecialization": []
604 }
605 }
606 }
607 }

```

Collection names

- provider-scdhhs-canonical
- provider-scdhhs-canonical-error

Parameter store Details

Parameter store name : **/mmis/lower/provider/in-ml-sql-glue**

Secret manager configuration

Following are the secrets in the secret manager

Secretkey	Secretvalue
writeUsername	Docdb write Username

writePassword	Docdb write password
host	Docdb host
port	Docdb port
ssl	Docdb ssl
canonicalDatabase	Final Canonical Model database name
rawDatabase	RDL data base name

S3 Bucket configuration

S3 bucket name : scdhhs-mmis-dev-lower

folder structure :

- under scdhhs-mmis-dev-lower **dev** folder
- under dev folder **code** and **data** folders
- under code folder **lib** , **src**, **tmp** folders
- under data folder **in** and **archive** folders

Email Notifications

Lambda Notifications

Note1: We will send notification only when error occurs.

Note2: To test this scenario, copy 8 files at once to “mftin” folder and then copy the 9th file to that location. To avoid multiple notification emails since Lambda will get triggered for each file as it is concurrency process.

Lambda Name: provider-mmis-metadata-in-cft

▼ Generic error for this particular Lambda

Lambda: blue-provider-mmis-metadata-in-cft Execution Details

Execution Status	ERROR
Start Time	2024-10-03 15:57:14
End Time	2024-10-03 15:59:04
Duration	1 minute and 50 seconds

Outcome	Failed to copy provider files from one location to another location. Further details available in CloudWatch.
----------------	--

Lambda Name: provider-mmis-in-cft

Configuration Error Scenario

Lambda: blue-provider-mmis-in-cft Execution Details

Execution Status	ERROR
Start Time	2024-10-03 10:33:41
End Time	2024-10-03 10:33:43
Duration	0 minute and 2 seconds
Outcome	Internal Error in lambda function initiation. Further details available in CloudWatch.

Generic Error Scenario

Lambda: blue-provider-mmis-in-cft Execution Details

Execution Status	ERROR
Start Time	2024-10-03 11:16:55
End Time	2024-10-03 11:19:01
Duration	2 minutes and 5 seconds
Outcome	Failed to invoke provider Glue Job. Further details available in CloudWatch.

Glue Job Notifications

Success Scenario

Glue: blue-provider-mmis-in-cft Execution Details

Execution Status	SUCCESS
Start Time	2024-10-03 11:37:27
End Time	2024-10-03 11:48:13

Duration	10 minutes and 46 seconds
Outcome	Received: 68953 Succeeded: 68953 Failed: 0

▼ Partial Success Scenario

Glue: blue-provider-mmis-in-cft Execution Details

Execution Status	PARTIAL SUCCESS
Start Time	2024-10-03 13:17:57
End Time	2024-10-03 13:29:36
Duration	11 minutes and 39 seconds
Outcome	Received: 68949 Succeeded: 68948 Failed: 1

▼ All Failed Records Scenario

Glue: blue-provider-mmis-in-cft Execution Details

Execution Status	ERROR
Start Time	2024-10-04 11:34:06
End Time	2024-10-04 11:45:08
Duration	11 minutes and 1 second
Outcome	Error occurred while mapping and got 68953 failed records. Further details available in CloudWatch.

▼ Error occurred while writing to document DB Scenario

Glue: blue-provider-mmis-in-cft Execution Details

Execution Status	ERROR
-------------------------	-------

Start Time	2024-10-04 10:31:23
End Time	2024-10-04 10:37:49
Duration	6 minutes and 26 seconds
Outcome	Error occurred while writing to document DB. Further details available in CloudWatch.

NOTE: For all configuration errors, we will send same notification message

▼ Configuration Error Scenario such as parameter store.

Glue: blue-provider-mmis-in-cft Execution Details

Execution Status	ERROR
Start Time	2024-10-03 12:05:00
End Time	2024-10-03 12:05:02
Duration	0 minute and 2 seconds
Outcome	Internal Error occurred while initializing provider glue job config. Further details available in CloudWatch.

▼ Generic Error scenario

Glue: blue-provider-mmis-in-cft Execution Details

Execution Status	ERROR
Start Time	2024-10-03 12:27:42
End Time	2024-10-03 12:38:20
Duration	10 minutes and 37 seconds
Outcome	Provider Batchjob failed while processing MMIS provider files. Further details available in CloudWatch.

Notes

While comparing documents with Marklogic and DocumentDb, the RDL records need to be deleted from the Marklogic collection for each load. If we don't delete RDL documents in ML, we get all the documents to TDS from Marklogic, whereas DocumentDB runs as a fresh load for each load.

- Is fee collected is changed from Boolean to string in enrollment
- ProvSpeciality is changed to ProviderSpeciality
- The Sort functionality is not working properly in Marklogic which is working correctly now
- The contact object structure has been changed according to architecture decision
- Facilites object in Marklogic is not populating correctly in location whereas it's populating correctly in docdb
- In address name key is not populating in Marklogic now its populating correctly
- All the empty objects of address is populating in locations now in the new code the empty object in location will be removed
- In marklogic "isprimary" in specialty is not populating correctly now it is populating
- Specialty code is added in summary of specialization
- Triples, mapper and publicID will not be present in DocumentDB
- In atypical provider documents under provider location, controlFacility should be populated to all addressTypes which is not populating the marklogic

Updates As part of 5.78.0:

- There were inconsistencies in the spelling of the attributes providerSpecialty and specialityCode. These have been corrected to ensure uniformity across all instances
- Updated the metadata to align with standards and removed the NPI from headers, as well as deleted metadata from headers.
- Updated the DocumentDB ingestion logic based on ProviderType. If the type is Person, the Organization block will not be shown, and vice versa.
- Other bug fixes as mentioned In the PR.